# MILITARY SPECIFICATION

## LIGHT, MARKER, DISTRESS, BUOYANT, DROPPABLE

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

## 1. SCOPE

1.1 This specification covers the requirements for the procurement of one type of a battery powered droppable buoyant waterproofed distress marker light.

## 2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings form a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply:

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-636 | Box, Fiberboard |
| PPP-T-45 | Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing |

Military

| | |
|---|---|
| MIL-P-116 | Preservation, Method of |
| MIL-T-5422 | Testing, Environmental, Aircraft Electronic Equipment |

STANDARDS

Federal

| | |
|---|---|
| Fed. Test Method Std. No. 151 | Metals; Test Methods |

FSC 6230

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Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |
| MS27981 | Fasteners, Snap, Style 2A (Small Wire Spring Clamp Type) |

DRAWINGS

Naval Air Systems Command

| | |
|---|---|
| 65A97D1 | Droppable Buoyant Distress Marker Light Assembly |
| 65A97D2 | Body Assembly, Droppable Buoyant Distress Marker Light |
| 65A97D3 | Header Assembly, Droppable Buoyant Distress Marker Light |
| 65A97D4 | Header and Reflector, Droppable Buoyant Distress Marker Light |
| 65A97C5 | Switch Release Assembly, Droppable Buoyant Distress Marker Light |
| 65A97C6 | Nose Assembly, Droppable Buoyant Distress Marker Light |
| 65A97C7 | Lens, Droppable Buoyant Distress Marker Light |
| 65A97D8 | Socket Assembly, Droppable Buoyant Distress Marker Light |

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government quality control representative. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)



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## 3. REQUIREMENTS

3.1 Preproduction - Unless otherwise specified, the droppable buoyant distress marker lights furnished under this specification shall be a product which has been inspected and has passed the preproduction inspections specified in Section 4.

3.2 Material and components - Materials and components shall conform to applicable specifications and drawings as listed or required herein. Unless otherwise specified, all the elastomeric materials used in the construction of the droppable buoyant distress marker lights shall have been manufactured not more than 12 months prior to the date of delivery of the lights. The batteries and lamps shall be new (never used). The batteries shall be not more than six months old.

3.3 Design - The light shall be a droppable, buoyant, waterproofed unit consisting essentially of a case, a nose filled with lead shot, a power supply, one lamp, one lens, and a push pin spring-loaded switch. The arrangement of the components shall conform to 65A97D1.

3.4 Construction - The light shall be constructed in accordance with 65A97D1 through 65A97D4, 65A97C5 through 65A97C7, and 65A97D8.

3.4.1 Power supply - The power supply shall consist of five manganese alkaline "D" cell batteries connected in series to provide a minimum output of 6.7 volts, when inspected as specified in 4.6.1. The batteries shall be enclosed in a waterproof tube fabricated from kraft paper, plastic, fabric base, or a combination thereof. The visible positive and negative terminals shall be the following snap fasteners:

| TERMINAL | POSITIVE | NEGATIVE |
|---|---|---|
| Part No. | MS27981-3N | MS27981-4N |

The snap fasteners, as applicable, shall mate with the snap fasteners on the wires of the header and socket assemblies, 65A97D3 and 65A97D8. The overall length and diameter of the assembled power supply shall be 12-3/8 by 1-3/8 ±1/32 inches. The assembled battery power supply shall be Eveready Division of Union Carbide Corporation, New York, New York, Part No. Y-1230, or equivalent.

3.5 Color - The color of the various components of the light shall be as specified in the applicable drawings.

3.6 Performance inspections -

3.6.1 Light intensity - When inspected as specified in 4.6.3 through 4.6.3.3, the minimum average light intensity (candlepower) of the horizontal, median, and vertical beams shall be as follows:



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| Light Beam | Minimum Average Candlepower |
|---|---|
| Horizontal | 10 |
| Median | 10 |
| Vertical | 150 |

3.6.2 Expansion due to altitude and waterproofness - When inspected as specified in 4.6.4, the case or any component shall not increase in volume, burst, crack, deform, or leak. The case or any component shall not separate and the lens shall not cloud or fog. All the components shall remain perfectly intact. The light shall then be inspected and conform to the requirements for light intensity, 3.6.1.

3.6.3 Switch activation - When inspected as specified in 4.6.5, the switch shall function properly without any difficulty.

3.6.4 Exposure cycle - When inspected as specified in 4.6.6, the case or any component shall not burst, crack, or deform. The case or any component shall not separate and the lens shall not cloud or fog. The light shall then be inspected and conform to the requirements for light intensity, 3.6.1, and switch activation, 3.6. The exposure cycle inspection shall be performed in sequence with the humidity inspection. The exposure cycle inspection shall be performed first and when completed the light shall then be inspected for conformance to the requirements for humidity, 3.6

3.6.5 Humidity - When inspected as specified in 4.6.7, the case or any component shall not burst, crack, deform, or separate. The lens shall not cloud or fog. The light shall then be inspected and conform to the requirements for light intensity, 3.6.1, and switch activation, 3.6.3.

3.6.6 Vibration - When inspected as specified in 4.6.8, the light shall not be lighted and the case or any component shall not crack, deform, or separate. The light shall then be inspected and conform to the requirements for light intensity, 3.6.1, and switch activation, 3.6.3.

3.6.7 Salt spray - When inspected as specified in 4.6.9, the light shall not leak or show any evidence of corrosion. The case or any component shall not crack, deform, or separate. The lens shall not cloud or fog. The light shall then be inspected and conform to the requirements for light intensity, 3.6.1, and switch activation, 3.6.3.

3.6.8 Impact - When inspected as specified in 4.6.10, the light shall float with the lens above the surface of the water and the light shall remain lighted. The light shall not leak and the case or any component shall not burst, crack, break, deform, or separate. The lens shall not cloud or fog. The light shall then be inspected and conform to the requirements for light intensity, 3.6.1.



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3.6.9 Weight - When weighed as specified in 4.6.11, the light minus the power supply but including the lamp shall weigh not more than 1.6 pounds. The power supply shall weigh not more than 1.6 pounds.

3.7 Markings -

3.7.1 Light - Each light shall contain the markings and be approximately positioned in the location specified in 65A97D4. The markings shall be hot stamped onto the header and all the letters and numerals shall be legible.

3.7.2 Power supply - Unless otherwise specified, the markings shall be legible and durable black letters and numerals which shall be thoroughly dry prior to the installation within the light. The marking ink shall not contain any material which is deleterious to the tube or power supply. The tube of the assembled power supply shall contain the following markings:

POWER SUPPLY
DROPPABLE BUOYANT DISTRESS MARKER LIGHT
MIL-L-81509(AS)
DATE OF MANUFACTURE (MONTH AND YEAR)

3.8 Dimensions - The dimensions of the light shall conform to the applicable drawings. The dimensions of the power supply shall conform to 3.4.1.

3.9 Workmanship - The droppable buoyant distress marker light shall not be misaligned, deformed, nor contain any rough surface, nicks, burrs, sharp edges, cracks, embedded foreign matter, detached components, or loose particles. The light shall be uniform in quality and shall be free from mechanical, electrical, or other irregularities or defects which could adversely affect performance, reliability or durability. The lights shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptance criteria established herein.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The examination and testing of the lights shall be classified as follows:



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(a) Preproduction inspection - Preproduction inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification.

(b) Quality conformance inspection - Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification.

4.3 Preproduction inspection - The preproduction inspection of the lights shall consist of examinations and tests for all of the requirements of this specification.

4.3.1 Preproduction samples - Unless otherwise specified, as soon as practicable after the award of the contract or order, the manufacturer shall submit five lights and five additional power supplies. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a manufacturer is in continuous production of these lights from contract to contract, submission of further preproduction samples on the new contract may be waived at the discretion of the procuring activity. Approval of the preproduction samples or the waiving of the preproduction inspection does not preclude the requirements for performing the quality conformance inspection. The preproduction samples shall be forwarded as specified in the contract or order (see 6.2). The samples shall be plainly identified by securely attached durable tags marked with the following information:

Sample submitted by (name) (date) for preproduction inspection in accordance with the requirements of MIL-L-81509(AS) under Contract No.

4.3.1.1 Upon completion of the preproduction inspection, all the applicable inspection reports and, when applicable, recommendations and comments pertinent for use in monitoring production will be forwarded to the Government quality control representative. The lights and the additional power supplies will be consumed or destroyed in the preproduction inspection and shall not be considered as part of the quantity to be delivered under contract.

4.4 Quality conformance inspection - The sampling and inspection levels shall conform to MIL-STD-105. The quality conformance inspection shall consist of the following:



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Power supply voltage
Visual examination of the lights
Dimensional check of the lights
Light intensity
Expansion due to altitude and waterproofness
Switch activation
Weight of the light and battery
Preparation for delivery

4.4.1 Sampling -

4.4.1.1 Inspection lot -

4.4.1.1.1 Power supply - An inspection lot size shall be expressed in units of one power supply and shall consist of all the power supplies received by the light manufacturer at one time. The sample unit shall be one power supply.

4.4.1.1.2 Lights - An inspection lot size shall be expressed in units of one light (with the power supply and lamp installed) made essentially under the same conditions and from the same materials and components. The sample unit shall be one light (with the power supply and lamp installed).

4.4.1.1.3 Preparation for delivery - An inspection lot size shall be expressed in units of one fully prepared shipping container, containing lights with the power supplies and lamps installed, fully prepared for delivery from essentially the same materials and components. The sample unit shall be one shipping container, containing lights with the power supplies and lamps installed, fully prepared for delivery with the exception that it need not be sealed.

4.4.1.2 Sampling for tests and examinations - The sample size, acceptance criteria, tests, and examinations required for the lights shall be as specified in Table I.

4.5 Inspection conditions -

4.5.1 Atmospheric conditions - Unless otherwise specified, all the inspections required by this specification shall be conducted at an atmospheric pressure of 28 to 32 inches of mercury, at a temperature of 77 ±18 degrees Fahrenheit (25 ±10 degrees Centigrade), and a relative humidity of 80 percent or less. When the inspections are conducted at an atmospheric pressure or temperature different from the above values, proper corrections shall be made for the change in instrument readings.



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TABLE I

SAMPLE SIZE, ACCEPTANCE CRITERIA, TESTS, AND EXAMINATIONS OF THE LIGHTS

| INSPECTION | CLASSIFICATION OF CHARACTERISTIC | PARAGRAPH | | SAMPLE SIZE | ACCEPTANCE CRITERIA |
|---|---|---|---|---|---|
| | | REQUIREMENT | METHOD | | |
| Voltage (Power Supply) | Major | 3.4.1 | 4.6.1 | Every power supply | Reject all defective units |
| Visual Examination | Major and Minor | Section 3 and applicable drawings | 4.6.2.1 | Every light for major defects and Inspection Level II for minor defects | Reject all units with any major defect and an acceptable quality level of 6.5 defects per 100 units for minor defects |
| Dimensional Check | Major | 3.8 and applicable drawings | 4.6.2.1 | Inspection Level S-3 | An acceptable quality level of 6.5 defects per 100 units |
| Light Intensity | Major | 3.6.1 | 4.6.3 through 4.6.3.3 | Inspection Level II 1/ | Acceptance number zero, rejection number 1 |
| Expansion Due to Altitude and Waterproofness | Major | 3.6.2 | 4.6.4 | Inspection Level S-2 1/ | Acceptance number zero, rejection number 1 |
| Switch Activation | Major | 3.6.3 | 4.6.5 | Inspection Level S-3 1/ | Acceptance number zero, rejection number 1 |
| Weight | Major | 3.6.9 | 4.6.11 | Inspection Level S-3 1/ | Acceptance number zero, rejection number 1 |
| Preparation for Delivery | Major | Section 5 | 4.6.2.2 | Inspection Level S-2 | An acceptable quality level of 2.5 percent defective |

1/ The sample size shall be based only on the applicable sample size code letter corresponding to the specified inspection level of MIL-STD-105.



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4.5.2 Synthetic sea water - When sea water is required in an inspection method, synthetic sea water of the composition specified in Table II shall be used. The synthetic sea water shall be prepared by dissolving the sodium sulfate in distilled water and then adding to the solution containing the other dissolved salts. If necessary, an adjustment of the pH to 8.0 shall be made by the addition of sodium hydroxide. The final volume shall be brought up to one liter.

TABLE II

COMPOSITION OF THE SYNTHETIC SEA WATER

| INGREDIENTS | GRAMS PER LITER |
|---|---|
| Anhydrous Sodium Chloride (NaCl) | 22.347 |
| Magnesium Chloride ($MgCl_2.6H_2O$) | 9.759 |
| Anhydrous Sodium Sulfate ($Na_2SO_4$) | 3.337 |
| Anhydrous Calcium Chloride ($CaCl_2$) | 0.944 |
| Anhydrous Sodium Bicarbonate ($NAHCO_3$) | 0.168 |

4.6 Inspection methods -

4.6.1 Voltage - The voltage at the power supply terminals shall be determined with a voltmeter, while discharging through a load of 15 ohms. The indicator needle shall deflect a minimum of 25 percent of the voltmeter scale.

4.6.2 Visual examination -

4.6.2.1 Lights - Every light shall be examined visually for major defects to determine conformance to this specification. Each light, selected as a sample unit from the lot, shall be thoroughly checked dimensionally and examined visually for minor defects to determine conformance to this specification. The classification and list of defects, Tables III and IV, shall be used to classify and enumerate the defects found.

4.6.2.2 Preparation for delivery - Each of the fully prepared shipping containers, containing lights with the power supplies and lamps installed, selected as a sample unit from the lot, shall be visually examined to determine that the packaging, packing and marking conform to this specification. The list of defects, Table V, shall be used to enumerate the defects found.



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TABLE III

CLASSIFICATION OF DEFECTS FOR THE VISUAL EXAMINATION OF THE LIGHTS

| MAJOR | MINOR |
|---|---|
| 101. Any surface rough, misaligned, or contains nicks, sharp edges, cracks, burrs, embedded foreign matter, loose particles, or other flaws. | 201. Any identification marking missing, incorrect, illegible, or incomplete. |
| 102. Any finish missing or any component improperly finished. | 202 Color of any component not as specified. |
| 103. Any component missing, malformed, corroded, fractured, or otherwise damaged. | |
| 104. Any component loose, detached, not attached as specified, or otherwise not securely retained. | |
| 105. Any power supply or lamp not installed within the light. | |
| 106. Any functioning part that works with difficulty. | |
| 107. Any spot or stain. 1/ | |

1/ The defect shall be considered major if it affects the serviceability, otherwise it is to be classified as a minor defect.

TABLE IV

LIST OF DEFECTS FOR FINISHED DIMENSIONS OF THE LIGHTS

| EXAMINE | DEFECT |
|---|---|
| Measure the individual components without damaging the assembled light. | Any measurement deviating from the dimensions and tolerances as specified in the specification and applicable drawings shall be enumerated as a dimensional defect. |



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TABLE V

LIST OF DEFECTS FOR PREPARATION FOR DELIVERY

| ITEM | DEFECT |
|---|---|
| Exterior and interior markings | Missing, incorrect, incomplete, illegible; of improper size, location, sequence, or method of application; markings not the same on the interior and exterior containers. |
| Packaging and packing materials | Any non-conforming component; any component missing, damaged, or otherwise defective. |
| Workmanship | Inadequate application of the components such as incomplete closure of the unit package, container flaps, loose strappings, etc.; bulging or distortion of the containers. |
| Exterior and interior weight or content | Gross or net weight exceeds the requirement. Light and power supply packaged separately. |

4.6.3 Light intensity - The equipment for determining the light intensity, candlepower of the horizontal, median, and vertical beams of the light, shall consist of a rotating and tilting apparatus and a calibrated photometer.

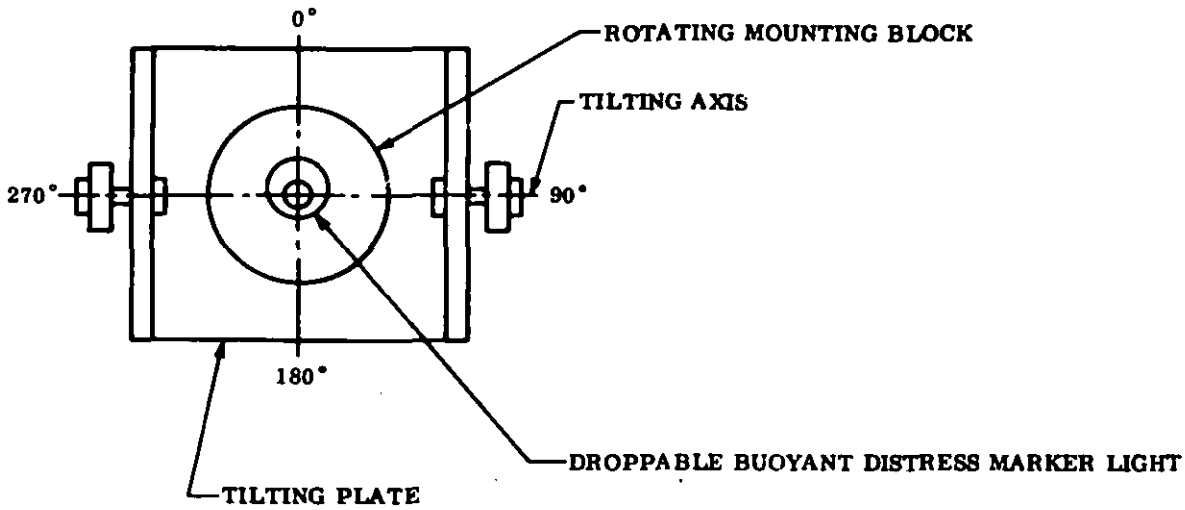
4.6.3.1 Equipment -

4.6.3.1.1 Rotating and tilting apparatus - The apparatus shall consist of a tilting plate and a rotating mounting block. The rotating mounting block shall be designed to securely hold the droppable buoyant distress marker light and shall be capable of rotating the distress marker light around its center axis, perpendicular to the plate surface, independently of the plate orientation. The mounting block shall be secured to the plate in such a manner that the tilting axis of the plate is not more than 1/4 inch above or below the lamp filament. The rotating and tilting apparatus shall conform to Figure 1.

4.6.3.1.2 Photometer - The photometer used shall be calibrated and shall be capable of measuring the candlepower, of the various light beams, in the ranges specified in 3.6.1. The deflection of the photometer indicator shall be not less than 25 percent of full scale for any light beam reading. The spectral response of the photometer light sensing element shall approximate the CEI



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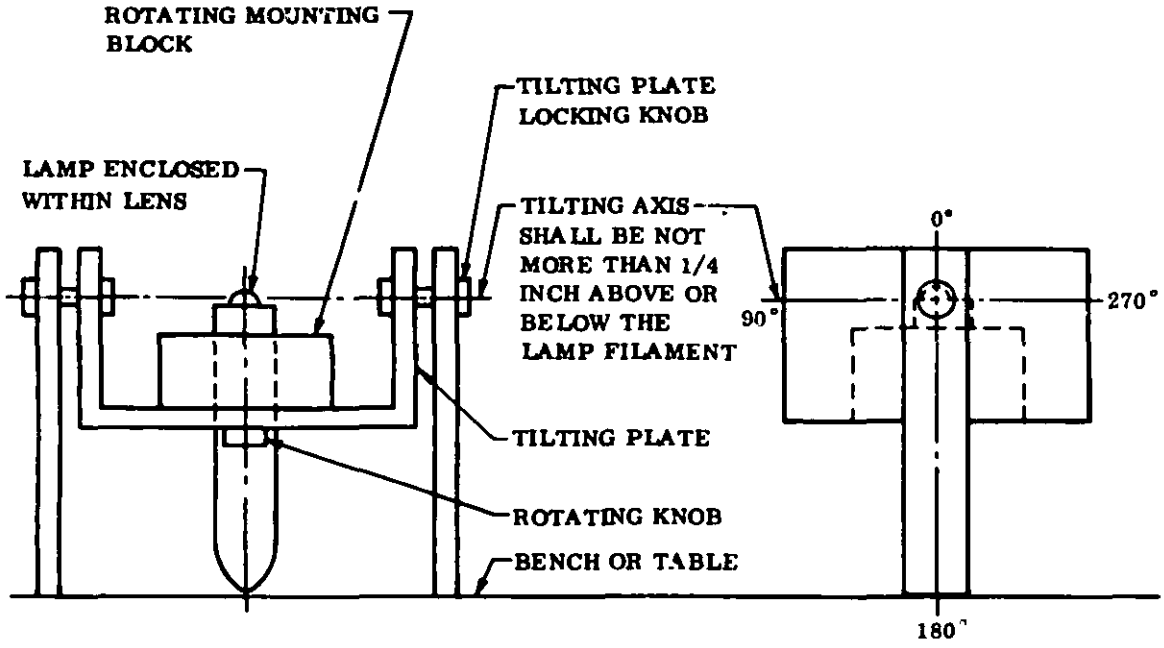




FIGURE 1. LIGHT INTENSITY ROTATING AND TILTING APPARATUS



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luminous efficiency curve for the spectral response of the human eye. The accuracy of the photometer shall be verified, prior to the start of each sample lot inspection, by comparison with a standard lamp, which has been aged and calibrated in accordance with accepted photometric practice, by a recognized established authority.

4.6.3.2 Procedure - The light to be inspected shall contain the power supply and lamp. The rotating and tilting apparatus and the photometer shall be placed on a level, firm, stationary bench or table. The lamp of the droppable buoyant distress marker light and the photometer light sensing element shall be a minimum distance of two feet apart. The lamp filament of the distress marker light shall be positioned parallel to the plane of the photometer light sensing element. All the light intensity inspections shall be conducted in a dark room environment. Prior to the start of each individual inspection, the photometer indicator shall be checked and adjusted to zero, if necessary. The light shall be securely attached to the rotating and tilting apparatus and activated. The candlepower of each of the horizontal and median annular light beams shall be determined at four positions around the beam, 0, 90, 180, and 270 degrees. The 0 degree is defined as the azimuth position in which an imaginary line, between the photometer light sensing element and the lamp filament of the droppable buoyant distress marker light, intersects the long axis of the lamp filament at an angle of 90 degrees. Only one determination for the candlepower of the vertical light beam shall be made. The rotating and tilting apparatus shall be tilted, as required, to superimpose the maximum intensity of the light beam directly upon the photometer light sensing element. The maximum intensity of the light beam reading, at each of the four positions, shall be recorded.

4.6.3.3 Report - The candlepower for the horizontal and median light beams shall be the average of the four maximum intensity candlepower readings for each light beam reported to the nearest 0.1 of a candlepower. The candlepower for the vertical beam shall be the individual maximum intensity reading obtained and reported to the nearest 0.1 of a candlepower.

4.6.4 Expansion due to altitude and waterproofness - The light to be inspected shall contain the power supply and lamp. The light shall be placed in a suitable air-tight container. The pressure within the container shall be reduced to an absolute pressure of 75 ±10 millimeters of mercury in a maximum time of 15 minutes. The pressure shall be maintained for 5 ±1 minutes and the light observed for increase in volume, bursting, cracking, deformation, separation of the components and for fogging or clouding of the lens. Upon completion of the observation, the pressure in the air-tight container shall be returned to normal within one minute and the light examined as aforementioned. Upon completion of the examination and without making any adjustments, the light shall be submerged in the synthetic sea water at a temperature of 77 ±2 degrees Fahrenheit (25 ±1 degrees Centigrade) under a static head of 12 inches of water for 2 ±1/4 hours. The light shall be



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submerged in a manner that shall permit the water to circulate completely around the light. Upon the expiration of the submersion period, the light shall be removed from the water and the outside surface wiped dry. The light shall then be immediately subjected to the light intensity inspection, 4.6.3 through 4.6.3.3, without removing the lens and lamp or making any adjustment. After the candlepower for each of the light beams has been determined, the lens and nose shall be removed and the lens, header assembly, and the inside of the light examined for moisture, separation of the components, and bursting, cracking, deformation, and increase in volume of any component. Any evidence of moisture within the lens, header assembly, or inside the light is an indication that the light leaks and is not waterproof.

4.6.5 Switch activation - The light, with the power supply and lamp installed and with the switch assembly in the off position, shall be held in the right hand and the thumb shall be in contact with the rod of the switch release assembly. Thumb pressure shall be applied and the rod shall be released from the plate and at the same time shall release the switch pin. The light shall operate. The rod shall then be replaced, turning the light off by depressing the switch pin, and held in position by the switch release assembly. The aforementioned shall be repeated for a total of 10 times. There shall be no difficulty in the releasing or securing of the rod and switch pin.

4.6.6 Exposure cycle - The light to be inspected shall contain the power supply and lamp. The light shall be subjected, in the sequence listed, to the following conditions:

2 hours at minus 30 ±5 degrees Fahrenheit (minus 34 ±2.8 degrees Centigrade).
2 hours at 75 ±5 degrees Fahrenheit (24 ±2.8 degrees Centigrade).
2 hours at 160 ±5 degrees Fahrenheit (71 ±2.8 degrees Centigrade).
12 to 18 hours at 75 ±5 degrees Fahrenheit.

These conditions shall be repeated for a total of five cycles. The amount of elapsed time between each condition and cycle shall be not more than 30 minutes. The light shall be exposed in such manner that the conditioned air circulates completely around the light. Upon completion of the temperature cycles, the light shall be subjected to the light intensity inspection, 4.6.3 through 4.6.3.3, without removing the lens and lamp or making any adjustment. After the candlepower for each of the light beams has been determined, the lens and nose cone shall be removed. The light shall then be examined for cracks, breaks, deformation, separation of the components, and for fogging or clouding of the lens. Upon completion of the examination, the light shall be reassembled and subjected to the switch activation inspection, 4.6.5. Upon completion of the testing and examination, the light, containing the same power supply and lamp, shall be subjected to the humidity inspection, 4.6.7.



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4.6.7 Humidity - The light, containing the power supply and lamp, which was previously subjected to the exposure cycle inspection, 4.6.6, shall be exposed to humidity as specified in MIL-T-5422, except that the preconditioning is not applicable. The light shall be exposed in such a manner that the conditioned air and humidity circulate completely around the light. Upon completion of the humidity exposure, the outer surface of the light shall be wiped dry and immediately subjected to the light intensity inspection, 4.6.3 through 4.6.3.3, without removing the lens and lamp or making any adjustment. After the candlepower for each of the light beams has been determined, the lens and nose cone shall be removed. The light shall then be examined for cracks, bursting, deformation, separation of the components, and for fogging or clouding of the lens. Upon completion of the examination, the light shall be reassembled and subjected to the switch activation inspection, 4.6.5.

4.6.8 Vibration - The light to be inspected shall contain the power supply and lamp. The light shall be mounted on a vibration inspection stand and vibrated at a double amplitude of 0.03 ±0.006 inch. The frequency shall be varied uniformly from 10 to 55 cycles per second and returned to 10 cycles per second in approximately one minute cycles. The vibration shall be applied for a period of 1 hour in each of three mutually perpendicular planes. The light shall not have been activated (lighted) while being vibrated. Upon completion of the vibration and without making any adjustment, the light shall be subjected to the light intensity inspection, 4.6.3 through 4.6.3.3, without removing the lens and lamp. After the candlepower for each of the light beams has been determined, the lens and nose cone shall be removed. The light shall then be examined for cracks, breaks, deformation, and separation of the components. Upon completion of the examination the light shall be reassembled and subjected to the switch activation inspection, 4.6.5.

4.6.9 Salt spray - The light to be inspected shall contain the power supply and lamp. The light shall be exposed to 96 hours of salt spray in accordance with Fed. Test Method Std. No. 151, Method 811. The light shall be exposed in such a manner that the salt atmosphere circulates completely around the light. At the expiration of the salt spray exposure, the outer surface of the light shall be wiped dry and immediately subjected to the light intensity inspection, 4.6.3 through 4.6.3.3, without removing the lens and lamp or making any adjustment. After the candlepower for each of the light beams has been determined, the lens and nose cone shall be removed. The light shall then be examined for bursting, cracks, deformation, corrosion, separation of the components, and for fogging or clouding of the lens. Upon completion of the examination, the light shall be reassembled and shall be subjected to the switch activation inspection, 4.6.5.

4.6.10 Impact - The power supply and lamp shall be installed in the light to be inspected. While activated, the light shall be dropped onto the water from an aircraft flying at 120 to 150 knots per hour and at an altitude of 250 to 300 feet. The light shall float with the lens above the surface of the water and the light shall still be lighted. The light shall be recovered from the water within one hour after



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dropping and shall be extinguished. The outer surface of the light shall be wiped dry and the light shall be subjected to the light intensity inspection, 4.6.3 through 4.6.3.3, without removing the lens and lamp or making any adjustment. After the candlepower for each of the light beams has been determined, the lens and nose cone shall be removed and the lens, header assembly, and the inside of the light examined for moisture. Any evidence of moisture within the lens. header assembly, or inside the light is an indication that the light leaks and is not waterproof. The light shall also be examined for cracks, breaks, deformation, and separation of the components.

4.6.11 Weight - The weight of the light minus the power supply but including the lamp shall be determined on any scale or balance capable of weighing to the nearest 0.01 of a pound. The weight of the power supply shall also be determined in the same manner.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging - Packaging shall be Level A, B, or C, as specified (see 6.2).

5.1.1 Level A - The light, with the power supply and lamp installed, shall be preserved and cushioned in accordance with MIL-P-116, Method III. No contact preservative is required. The light shall be packaged within a snug fitting fiberboard container conforming to PPP-B-636, Type CF or SF, Domestic Class, Variety SW, Grade 125. Twelve of the boxed lights shall be packaged within a snug fitting fiberboard container conforming to PPP-B-636, Type CF or SF, Domestic Class, Variety SW, Grade 275. The individual and outer containers shall be constructed and closed in accordance with the appendix to PPP-B-636. All the seams and joints shall be sealed with a minimum of 3 inch wide tape conforming to PPP-T-45, Type III, Grade A, B, or C.

5.1.2 Level C - The light, with the power supply and lamp installed, shall be packaged to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

5.2 Packing - Packing shall be Level A, B, or C, as specified (see 6.2). Shipping containers, insofar as possible, shall be uniform in size and shape and of minimum cube and tare weight.

5.2.1 Level A - Twenty-four lights (two containers), packaged as specified in 5.1, shall be packed as specified in 5.2.2, except that the fiberboard container shall be Weather Resistant Class, Variety SW, Grade V3c or V3s. Each container shall be constructed, closed, and reinforced with flat steel strapping or tape banding in accordance with the appendix to PPP-B-636.



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5.2.2 Level B - Twenty-four lights (two containers), packaged as specified in 5.1, shall be packed within a snug fitting fiberboard container, conforming to PPP-B-636, Type CF or SF, Domestic Class, Variety SW, Grade 275. Each container shall be constructed and closed in accordance with the appendix to PPP-B-636.

5.2.3 Level C - The packaged lights, which require packing for acceptance by the carrier, shall be packed within exterior type shipping containers in a manner that shall assure safe transportation at the lowest rate to the point of delivery. The shipment shall conform to the minimum requirements of the rules and regulations applicable to the mode of transportation selected.

5.3 Marking - The interior and exterior containers shall be marked in accordance with MIL-STD-129 and shall include the date of manufacture (month and year).

6. NOTES

6.1 Intended use - The droppable buoyant distress marker light is intended for use by aircraft for visual aid in night air-sea rescue.

6.2 Ordering data - Procurement documents shall specify the following:

(a) Title, number, and date of this specification.

(b) Whether preproduction inspection is waived (see 4.3.1).

(c) Name and address of the preproduction inspection laboratory (see 4.3.1).

(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Data - For the information of Contractors and Contracting Officers, any of the data specified in applicable documents listed in Section 2 of this specification, or referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NavWeps Form 4200/15 (Drawings, Lists, and Specifications Required) shall be attached where applicable.



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

MIL-L-81509(AS) LIGHT, MARKER, DISTRESS, BUOYANT, DROPPABLE

| ORGANIZATION (Of submitter) | CITY AND STATE |
|---|---|
| | |

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

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